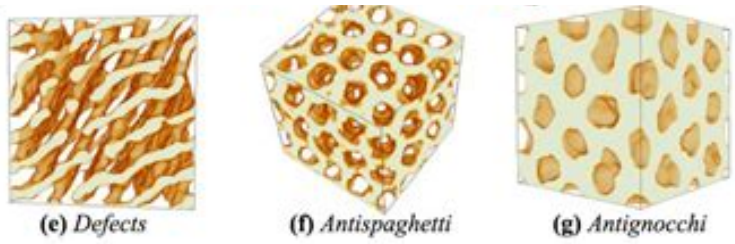


55-Cancri-Ae 💫 Design Doc

TARGET SHIP DATE 2026-04-24

Team Members: Michelle Chen, Natalie Keiger, Ethan Cheung, Thamidur Rahman

Project Name: Star-tling Discoveries

★ Overview

Dataset: Our dataset displays data on exoplanets drawn from the NASA Exoplanet Archive. The application guides visitors through a curated narrative tour of the dataset, illuminating findings and patterns that a casual visitor would not have been able to discover on their own. It can be used to understand exoplanet diversity, the history of discovery, and what these planets tell us about the stars they orbit.

Site: Our site will display the information from the dataset in an easily understandable table. The home page will guide the user, teaching them about what the data means. It will also teach them how to use the site. There will be a search bar if a user has a particular planet in mind. Each planet will have its own page which will include all the data about it and a simulation of its orbit. There will be a separate page to explain what the terms mean.

★ Goals of the Project

TEACH site users about exoplanets through guided tour on home page
DEMONSTRATE exoplanet orbits and characteristics through a search bar where users can find different exoplanets and filter them
COMPARE exoplanets using curated graphs, charts, and visuals so users can get an idea of what is out there!

★ General Project Deadlines

Task	Deadline
DD draft #1	3/26/26
Working app	TSD
Polished app	TSD
Final video uploaded (Meet in person)	TSD

★ Tasks

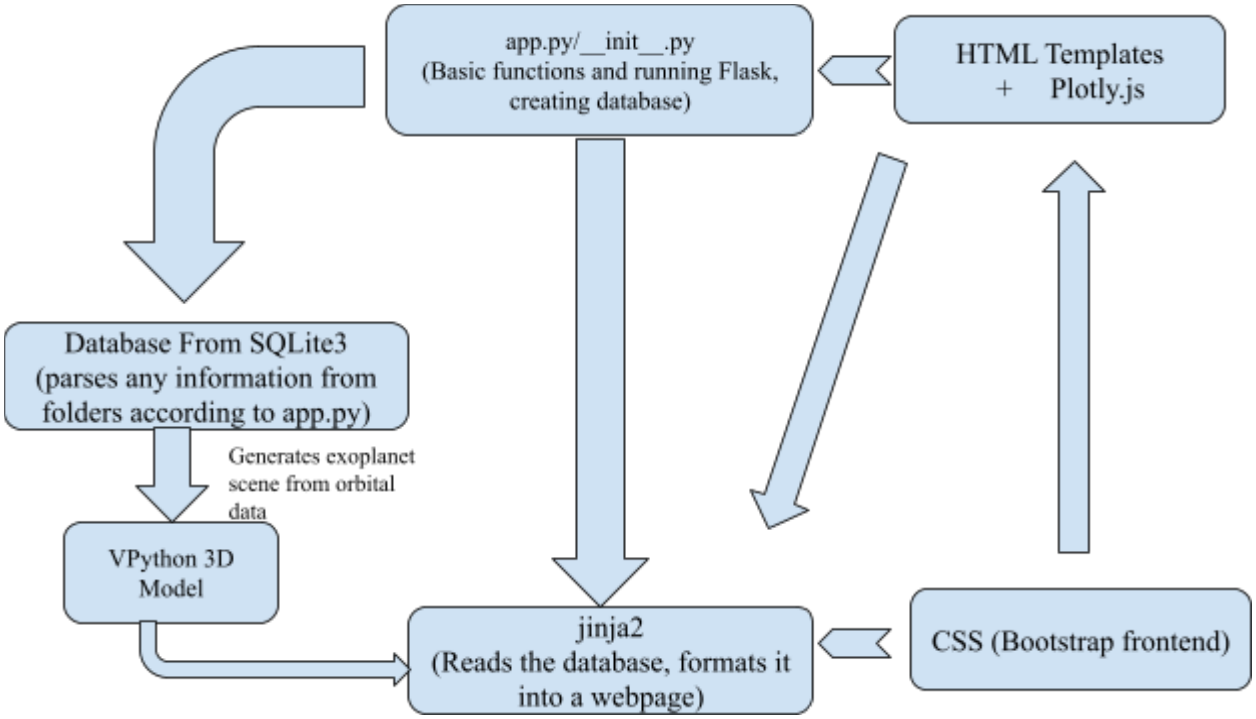
Task	Assigned members	Deadline	Status
Finish PC	Thamidur	3/26/26	100
Finish CM & DVL	Michelle	3/26/26	100
Finish SM	Ethan	3/26/26	100
Finish DBO	Natalie	3/26/26	100
Get webserver remotely working			0

★ Program Components

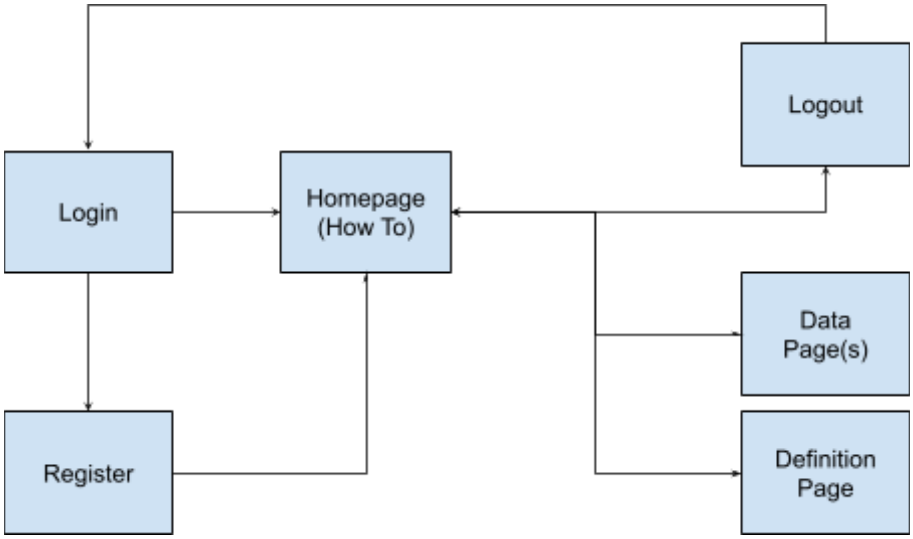
Component	Purpose
Frontend (HTML/CSS/JS + Bootstrap)	Renders all the pages and provides an interactive experience
Plotly Visualization	Responds to user filter input and used to display data in charts, graphs, and etc.

Flask Application	Renders templates and used for web server
VPython	Runs the simulation of the data, showing period and eccentricity
SQLite3 Database	Stores all the data from csv
Python Modules	Handles all data loading, query building, and CSV

★ Component Map



Site Map


Logout
Login
Homepage
(How To)
Data
Page(s)
Register
Definition
Page

★ Database Organization

Database framework: SQLITE3

Key	Value	Example	Units
name (PK)	TEXT	“WASP-14 b”	n/a
ra*	REAL	14.55176528	n/a
dec*	REAL	21.89469444	n/a
density	REAL	4.65633	g/cm^3
distance	REAL	160	pc
eccentricity*	REAL	0.091	n/a
mass	REAL	7.69234	M_j
smass*	REAL	1.31	M_{sun}
period	REAL	2.243752	days
binary	TEXT	“false”	n/a
disc_method	TEXT	“Transit”	n/a
inclination	REAL	84.32	n/a
radius	REAL	1.281	R_j
semi_axis	REAL	0.0367693	AU
disc_year	REAL	2009	year

*Explanation: RA – right ascension, one of two coordinates used to measure location; Dec – declination, the other coordinate used to measure location; Eccentricity: how ovoid the orbit shape is (0 is circle, 1 is line); Transit Time – amount of time it takes for the planet to cross its star; SMass – star mass; semi-axis: semimajor axis (dist from star)

★ Data Visualization Library

Plotly supports interactive charts and can be integrated with Flask via Jinja templates. The learning curve is low to medium and they have a diverse range of charts available.

Examples: Line chart showing how many exoplanet discoveries were in each year, bar graph of discoveries by location, etc.

★ Data Processing

We use three Python libraries to handle data processing and visualization: Pandas, SQLite, and Plotly. Pandas reads the CSV file downloaded from NASA's Exoplanet Archive and loads it into a dataframe. It provides an efficient way to filter, group, and manipulate a large amount of planet data. SQLite stores the loaded data in a local database file. When a page requests data, it is pulled from Sqlite back into a Pandas dataframe, which is then passed through to Plotly to generate visualizations. Plotly converts the dataframe columns into JSON, which is embedded into the HTML page and rendered in the browser as a graph.

★ Resources Used

Dataset: exoplanets (NASA exoplanet database)

Example website: ExoAnalytics (https://exoanalytics.fmacal.com/)

3D Modeling https://www.vpython.org/

Frontend framework: Bootstrap

APIs: no external APIs needed; The NASA Exoplanet Database will be downloaded as a CSV file and used in our SQLite3 database. Avoids runtime dependency on external APIs.